Tutorial - 05

1 - An Email Services provider wants to classify Emails as Spam or not Spam using a linear SVM classifier

Input features

→ no. of Suspicious words

→ no. of External links

→ Attachment present (0/1)

Suspicious words (x_1)	links (x_2)	Attachment (x_3)	class y
6	3	1	-1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1

Initial weights $w_1 = w_2 = w_3 = 1$; $b = 0$; $\eta = 1$

Sample 1: $(6, 3, 1)$ $\boxed{y = -1}$

$$\boxed{f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b}$$

$$f(x) = 6(1) + 3(1) + 1(1) + 0$$

$$= 10$$

$$y_i (f(x)) = y_i (f(x)) = (-1)(10) = -10$$

If $y\,f(x) \geq 1 \rightarrow$ perfect classification

If $y\,f(x) < 1 \rightarrow$ Mise classification

$\rightarrow$ Here, $y\,f(x) < 1$. So Mise classification

updating weights & bias

$$w_0^{new} = w_0^{old} + \eta \sum_{i=1}^{n} x_i y_i \quad \boxed{\eta = 1}$$

$$w_0^{new} = 0 + 1\,[(1)(-1)] \rightarrow -1$$

$$w_1^{new} = 1 - 1\,[1 - 1(-1)(6)] = -6$$

$$w_2^{new} = 1 - 1\,[1 - (1)(-1)(3)] \rightarrow -3$$

$$w_3^{new} = 1 - 1\,[1 - 1(1)(-1)] \rightarrow -1$$

$$(w_0, w_1, w_2, w_3) \rightarrow (-1, -6, -3, -1)$$

Sample 2 $(1, 0, 0)$ $\boxed{y = -1}$

$$f(x) = (-6)(1) + (-3)(0) + (-1)(0) + (-1) = -7$$

$$y\,(f(x)) = (-1)(-7) = 7$$

$$y\,(f(x)) \geq 1 \rightarrow \text{perfect classification}$$

updating weights

$$w_1^{new} = (-6) - (1)(-6) = 0$$

$$w_2^{new} = (-3) - (1)(-3) = 0$$

$$w_3^{new} = (-1) - (1)(-1) = 0$$

$$\boxed{w_0^{new} = w_0^{old}}$$

Sample 3: (4,2,1) $\boxed{y=+1}$

$(w_1, w_2, w_3, w_0) = (0,0,0,-1)$

$f(x) = 0\times 4 + 0(2) + 0(1) + (-1) = -1$

$y[f(x)] = (-1)(1) = -1$

$y[f(x)] < 1 \rightarrow$ Miss classification

updating weights and bias

$w_1^{new} = 0 - 1(0 - 1(1)4) = 4$

$w_2^{new} = 0 - 1[0 - 1(1)(2)] = 2$

$w_3^{new} = 0 - 1[0 - 1(1)1] = 1$

$w_0^{new} = -1 + 1(1) = 0$

Sample 4: (0,0,0) $\boxed{y=-1}$

$(w_1, w_2, w_3, w_0) = (4,2,1,0)$

$f(x) = 4(0) + 2(0) + 1(0) + 0 = 0$

$y[f(x)] = -1\times 0 = 0$

$y[f(x)] < 1 \rightarrow$ miss classification

updating weights & bias

$w_1^{new} = 4 - 1[4 - 1(-1)0] = 0$

$w_2^{new} = 2 - 1[2 - 1(-1)0] = 0$

$w_3^{new} = 1 - 1[1 - 1(-1)0] = 0$

$w_0^{new} = 0 + 1(-1) = -1$

Sample 5 (5,2,1)

$$f(x) = 0(5) + 0(2) + 0(1) + (-1) = -1$$

$y[f(x)] < 1 \longrightarrow$ miss classification

updating weights and bias

$$w_1^{new} = 0 - 1[0 - 1(1)5] = 5$$

$$w_2^{new} = 0 - 1[0 - 1(1)2] = 2$$

$$w_3^{new} = 0 - 1[0 - 1(1)1] = 1$$

$$w_0^{new} = -1 + 1(1)(1) = 0$$

Final weights & bias are

$w_1 = 5 \quad w_2 = 2 \quad w_3 = 1, \quad w_0 = 0$

$$\text{margin} = \frac{2}{\|w\|} \quad (\text{or}) \quad \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$= \frac{2}{\sqrt{(5)^2 + (2)^2 + (1)^2}} = \frac{2}{\sqrt{30}} \approx 0.365$$